

Money Problems Involving Percentages

Jonathan R. Bacolod

Percentage means “per hundred” and can be written as a fraction, decimal, or percent.

Discount and Sale Price

Discount is the amount deducted from the original price.

Sale Price is the price after the discount has been subtracted.

$$\text{Discount} = \text{Rate} \times \text{Original Price}$$

$$\text{Sale Price} = \text{Original Price} - \text{Discount}$$

Commission

Commission is the amount earned by a salesperson based on a percentage of sales.

$$\text{Commission} = \text{Rate} \times \text{Total Sales}$$

Sales Tax

Sales Tax is an additional amount charged on purchases and is usually expressed as a percentage of the selling price.

$$\text{Sales Tax} = \text{Rate} \times \text{Selling Price}$$

Simple Interest

Simple Interest is the amount earned or paid for the use of money over a period of time.

$$I = Prt$$

I is the interest, P is the principal, r is the annual interest rate as a decimal, and t is the time in years.

Examples

Example 1

A pair of shoes costs ₱1,800. It is sold at a 15% discount. Find the discount and sale price.

$$\text{Discount} = 0.15(1800) = \text{₱}270$$

$$\text{Sale Price} = 1800 - 270 = \text{₱}1,530$$

Example 2

A real estate agent sold a property worth ₱2,500,000 and earns a commission of 3%. Find the commission.

$$\text{Commission} = 0.03(2,500,000) = \text{₱}75,000$$

Example 3

Find the simple interest on ₱5,000 invested at 4% per year for 3 years.

$$I = Prt$$

$$I = 5000(0.04)(3) = \text{₱}600$$

Practice Exercises

Practice 1

A backpack originally costs ₱1,250. During a sale, a discount of 20% is offered. Find the sale price.

Practice 2

A customer bought a tablet worth ₱12,000. If the sales tax rate is 12%, find the amount of sales tax.

Practice 3

Find the simple interest on ₱8,500 borrowed at 5% per year for 2 years.

Practice 4

A jacket priced at ₱2,400 is discounted by 25%.
Find the discount and the sale price.

Practice 5

Ana invested ₱15,000 at 6% simple interest for 4 years. Find the interest earned.

Activities

Activity 1

A wristwatch costs ₱2,200. It is sold at a 10% discount. Find the discount and the sale price.

Activity 2

A school bag originally costs ₱1,500. During a sale, a discount of 18% is offered. Find the sale price.

Activity 3

A salesperson sold furniture worth ₱52,000 and earns a commission of 5%. How much commission does the salesperson receive?

Activity 4

A customer bought a laptop worth ₱18,000. If the sales tax rate is 12%, find the amount of sales tax.

Activity 5

Find the simple interest on ₱6,500 invested at 5% per year for 4 years.

Activity 6

Find the simple interest on ₱10,000 borrowed at 4% per year for 3 years.

Activity 7

A bicycle priced at ₱3,600 is discounted by 30%. Find the discount and the sale price.

Activity 8

Carlo invested ₱20,000 at 5% simple interest for 3 years. Find the interest earned.

Thank you!